Title 2:

Enhanced Accuracy Prediction in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with Gradient Boosting Algorithm

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Keywords: Extreme Gradient Boosting, Comparing, Gradient Boosting, Accuracy, Prediction, Inflation Rate.

Abstract:

Aim: The purpose of this study is to assess how well various machine learning algorithms perform in raising the precision of inflation rate forecasts. In particular, it evaluates the effectiveness of the Gradient Boosting and the XGBoost (extreme gradient booster) algorithm to ascertain which model gets the most accurate predictions. **Materials And Methods:** The study used an information set that included 15 years' worth of previous inflation rate statistics from 19 different nations. Financial measurements, modifications to policy, and other pertinent indicators of the economy are included in the dataset. Ten repetitions of every method were performed, with 10 instances of in every group. The exact same information and the same set of parameters were used to train both of these models. A true-positive score of 0.132 was attained. **Result:** The results show that the two methods' accuracy in forecasting differs significantly. With a success rate of 90.11%, the model using XGBoost outperformed the method known as Gradient Boosting technique, which had an efficiency of 78.08%. **Conclusion:** According to the study's findings, the XGBoost algorithm predicts inflation rates more precisely than the Gradient Boosting model. When it comes to precisely predicting inflation rates, XGBoost is the better option, with a level of accuracy of 90.11% as opposed to 78.08%. As a result, it is suggested as the best technique to forecast inflation rates.

Keywords: Accuracy, Comparison, Between, Gradient Boosting, Extreme Gradient Boosting, Prediction, Inflation Rate

Introduction:

To make effective economic decisions, businesses, politicians, and financial institutions need to be able to estimate inflation rates accurately. Advanced machine learning approaches have evolved to increase forecasting precision as global economies grow more complicated. Both Gradient Boosting (GB) and XGBoost, also known as are potent ensemble learning techniques that have shown impressive results in predictive modeling. Although both methods use boosting concepts, XGBoost adds optimization strategies to improve accuracy and performance. By comparing how well XGBoost and GB anticipate inflation rates, this study seeks to determine which algorithm provides the most accuracy and dependability(Safira et al. 2024).

This study's main goal is to use both algorithms to examine past inflation data and assess how well they anticipate future events. Gradient Boosting is a popular technique in economic modeling because of its reputation for reducing mistakes by gradually enhancing weak learners. Nevertheless, XGBoost offers other improvements including regularization and effective management of missing data, which might give inflation rate forecasting a competitive edge. This study aims to ascertain whether approach produces more accuracy and resilience in actual financial datasets by analyzing their prediction capacities(Sun, Yin, and Lu 2025).

Dealing with shifting market circumstances and a variety of economic factors that affect inflation patterns is one of the biggest problems in economic forecasting. To improve forecasting results and refine predictive models, it is essential to comprehend the relative advantages and disadvantages of GB and XGBoost. By offering a thorough assessment of both algorithms, this study seeks to close the gap in previous research and eventually aid in better decision-making in risk management, financial planning, and economic policy(Somakumar et al. 2025).

Materials And Methods:

Assessing the predictive capabilities of several artificial intelligence models in predicting inflation rates inside SIMATS was the aim of this study. The models were divided into two groups: the XGBoost (extreme gradient boosting) methodology was used in the second group, while the Gradient Increasing method was used in the first group. To establish a fair comparison, statistical computations were conducted with alpha = 0.05 & beta = 0.2, having the two models run at separate intervals

Inflation rate statistics from several nations over a 15-year period make up the dataset utilized in this study. Important economic indicators, financial measurements, and policy modifications that affect inflation patterns are all included. Frequent pre-processing procedures were used to improve the quality of the dataset, guaranteeing accurate and trustworthy analysis. To preserve consistency and get rid of inconsistencies, a variety of data cleaning strategies were used, which produced forecasts that were more accurate.

Extreme Gradient Boosting:

To improve model performance, the XGBoost technique uses an iterative boosting procedure. It incorporates regularization strategies and weighted updates during training to reduce mistakes and raise prediction accuracy. Better model accuracy results from this boosting mechanism's efficient reduction of bias and variation.

Among the many benefits of XGBoost are its remarkable ability to handle intricate relationships, computational effectiveness, regularization-based resistance to overfitting, and versatility with regard to big datasets. It also offers effective feature selection, which enhances the interpretability of the model. All things considered, XGBoost is well known for its excellent prediction accuracy, scalability, and resilience in a variety of forecasting workloads.

Gradient Boosting:

In order to improve overall model accuracy, the Gradient Boosting technique iteratively combines weak learners using a sequential boosting strategy. By reducing prediction mistakes, it optimizes performance through the step-by-step construction of decision trees.

The capacity of gradient boosting to increase model accuracy and capture nonlinear interactions is well-known. It offers feature significance determination, high generalization capabilities, and good performance across various data formats. To avoid overfitting, nevertheless, precise parameter adjustment is necessary. Although gradient booster is a good predictive model, it is not as accurate as XGBoost when it comes to predicting inflation rates.

Statistical Analysis:

IBM SPSS was used for quantitative analysis in order to evaluate the precision of inflation rate forecasting techniques. To assess the importance of the accuracy variances among the two models, techniques including cross-validation and hypothesis testing were employed. This study's goal is to conduct a thorough statistical analysis and identify the best strategy for improving the accuracy of inflation rate forecasts.

Result:

When it came to predicting inflation rates, the XGBoost (extreme gradient boosting) algorithm outperformed the Gradient Boosting method with an accuracy of 90.11%, whereas the latter method only managed 78.08%. As shown in Table 3, XGBoost continuously performed better than Gradient Boosting in predicting inflation rates across all rounds.

The statistical assessment of the two approaches is further demonstrated in Table 4. The stability and dependability of the XGBoost model were demonstrated by its mean performance of 90.11% with an acceptable standard variation (SD) of 0.084. On the other hand, the Gradient Boosting method performed quite poorly, with an average precision of 78.08% but a deviation from the mean (SD) of 0.00527. These findings confirm that the best method for increasing predicting accuracy for inflation rates is XGBoost.

Discussion:

According to the results of this research, the XGBoost (extreme gradient booster) method outperforms the conventional gradient boost (GB) technique in the field of price inflation forecasting, with a precision rate of 90.11% as opposed to 78.08%. A substantial difference in predicted accuracy was validated by a paired t-test, highlighting the benefit of XGBoost in reducing mistakes and identifying intricate economic patterns. Because of its improved regularization methods, capacity to tolerate missing data, and excellent computing efficiency, XGBoost performs better than other algorithms. These findings demonstrate that XGBoost is more dependable than conventional Gradient Boosting at predicting inflation patterns(Chaij et al. 2025).

Although GB and XGBoost both use boosting concepts, XGBoost improves generalization and speeds up training durations by including improvements like simultaneous processing as a more sophisticated method for dealing with poor learners. According to some research, classic gradient boosting is still competitive in smaller datasets because of its more straightforward structure, but XGBoost routinely performs better when working with large-scale economic data. The increasing agreement that XGBoost is a better option for financial forecasting, especially in situations where accuracy is crucial, is supported by this study(Xu et al. 2025).

These results highlight how crucial it is to choose the best learning model for predicting inflation. The study's findings are consistent with previous research that acknowledges the XGBoost as a more precise and scalable method of predicting economic trends. Further improvements in feature selection strategies, hyperparameter tweaking, and the incorporation of more macroeconomic data should all be investigated in future studies. A more thorough examination of these variables may result in forecasting models that are even more accurate, allowing financial experts and politicians to base their judgments on accurate inflation projections(Sioda et al. 2025).

Conclusion:

According to this study, the XGBoost (extreme gradient booster) algorithm performs better than the Gradient Boosting approach in terms of more accurate inflation rate predictions. The accuracy of the Gradient Boosting technique was just 78.08%, whereas XGBoost's precision was 90.11%. The findings show that XGBoost offers a more accurate and dependable method of

predicting inflation trends. For financial experts and politicians looking to create precise inflation rate forecasts, XGBoost is the best option due to its exceptional predictive accuracy. It is a useful tool when it comes to forecasting because of its capacity to manage intricate economic variables and policy changes.

Declaration:

There are no dual loyalties with this document.

Authors Contribution:

JSC had been in control of writing the manuscript, as well as data analysis and collection. Author KVK was in charge of conceptualizing, validating data, and critically evaluating the content.

Acknowledgements:

The authors thank the SIMATS engineering, SIMATS institute of medical and technical science for supplying the essential support to complete this study effectively.

Funding:

The ensuing institution's contributions allowed me to finish this research, and we are grateful for their assistance.

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Table 1: pseudocode for XGBoost algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries required to implement the XGBoost algorithm. 2. Put the dataset into the program you are using. 3. Divide the dataset into assessment and development sets. 4. Make an object called XGBoost. 5. Use the development set to train the classifier. 6. Forecast test data labels. 7. Examine and contrast the assessment and development sets. 8. Determine the precision percentage.

Table 2: pseudocode for gradient boosting algorithm.

Input: inflation rate prediction dataset
Output: improved accuracy for inflation rate prediction
1. Load the libraries required to implement the gradient boosting algorithm. 2. Put the dataset into the program you are using. 3. Divide the dataset into assessment and development sets. 4. Make an object called gradient boosting. 5. Use the development set to train the classifier. 6. Forecast test data labels. 7. Examine and contrast the assessment and development sets. 8. Determine the precision percentage.

Table 3: Accuracy values of 10 iterations:

Iteration number	xg-boost accuracy	Gb accuracy
1	90.07%	78.08%
2	90.21%	78.08%
3	90.21%	78.09%
4	90.15%	78.09%
5	90.01%	78.09%
6	90.02%	78.09%
7	90.01%	78.08%
8	90.16%	78.09%
9	90.21%	78.08%
10	90.10%	78.08%

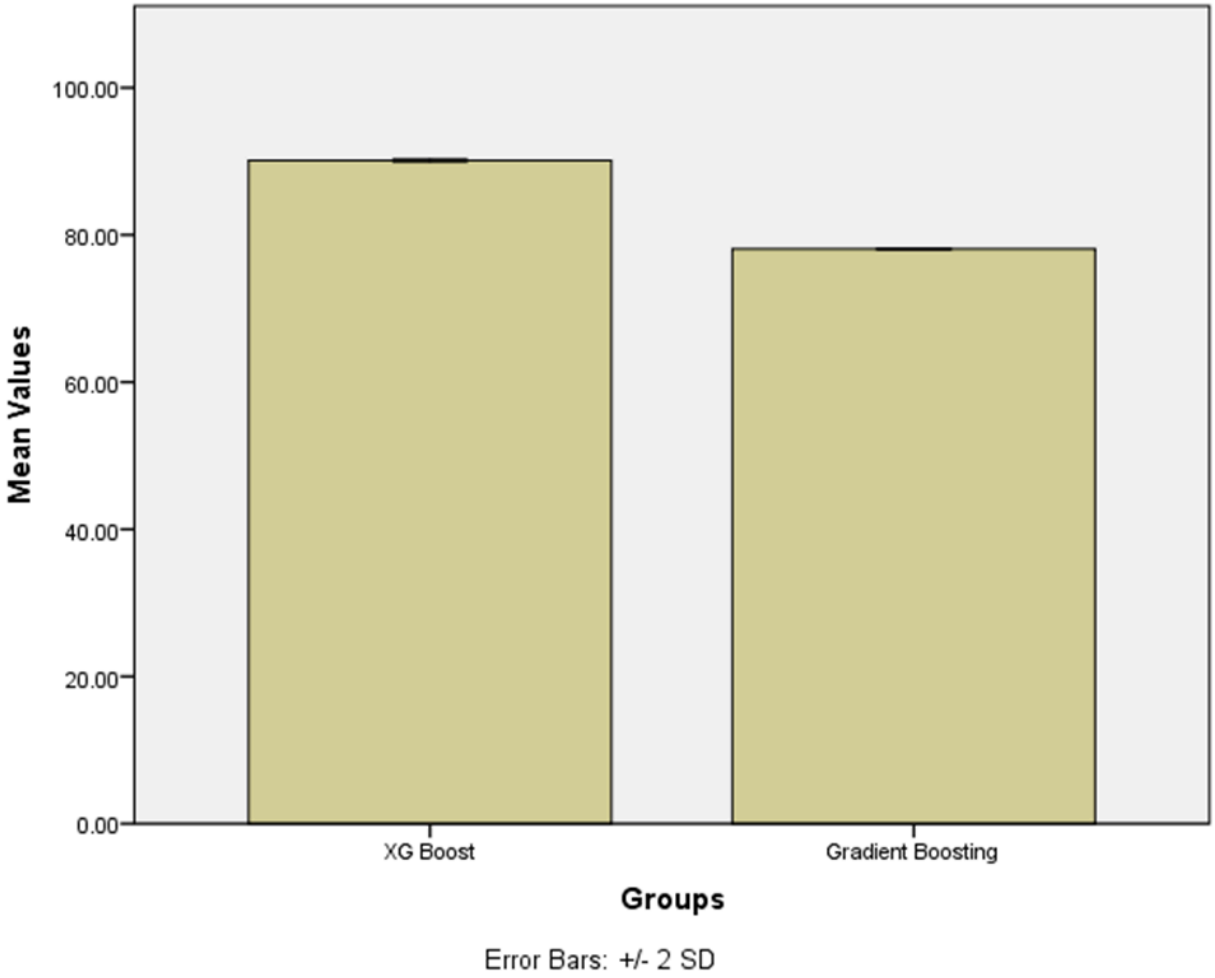
Table 4: Analysis of mean & SD

group	n	average	standard deviation	standard error mean
XGBoost	10	90.1150	0.08410	0.02659
GB	10	78.0850	0.00527	0.0167

Table 5: Independent Sample Test

independent samples test											
		levene's test for equality of variances		t-test for equality of means							
		f	sig.	t	df	sig. (2-tailed)	mean difference	std. error difference	95% confidence interval of the difference		
									lower	upper	
accuracy	equal variances assumed	0.52	0.121	451.478	18	0.00	12.03	0.02665	11.97	12.08	
	equal variances not assumed			451.478	9.071	0.00	12.03	0.02665	11.96	12.09	

Figure 1: Extreme Gradient Boosting algorithm **Vs** Gradient boosting algorithm



References:

Chaij, Jasmine M., Jacob B. Hammond, Skyler K. Palmer, Anna D. Lee, Diego A. Gomez, Brooke M. French, Phuong D. Nguyen, David W. Mathes, and David Y. Khechoyan. 2025. "Medicaid's Cranio-Cap: Medicaid Reimbursement for Craniosynostosis Repair Is Not Trending With the Rate of Inflation." *The Journal of Craniofacial Surgery*, March. https://doi.org/10.1097/SCS.0000000000011165.

Safira, Anisya, Riswanda Ayu Dhiya'ulhaq, Indah Fahmiyah, and Mohammad Ghani. 2024. "Spatial Impact on Inflation of Java Island Prediction Using Autoregressive Integrated Moving Average (ARIMA) and Generalized Space-Time ARIMA (GSTARIMA)." *MethodsX* 13 (December):102867.

Sioda, Natasha A., David Etzioni, Kenan Shawwaf, Zhi Ven Fong, Justin T. Brady, Kayla L. Haydon, Ga-Ram Han, Emily Thompson, and Irving Jorge. 2025. “Comparison of Laparoscopic Cholecystectomy Reimbursement Rate and Medical Malpractice Insurance Premium Cost Across the US 2013-2023.” *Journal of the American College of Surgeons*, March. https://doi.org/10.1097/XCS.0000000000001376.

Somakumar, Shwetha, Fathima Thashreefa Basheer, Vijayanarayana K, Vani Lakshmi R, Shyamasunder N. Bhat, Gabriel Sunil Rodrigues, Girish Menon R, Elstin Anbu Raj S, and Rajesh V. 2025. “Factors Affecting Readmission in Patients with Surgical Site Infection: A Graphical and Prediction Model-Based Approach.” *Surgical Infections* 26 (2): 63–70.

Sun, Lu, Zhixiang Yin, and Lin Lu. 2025. “ISLRWR: A Network Diffusion Algorithm for Drug-Target Interactions Prediction.” *PloS One* 20 (1): e0302281.

Xu, Ziqi, Arya Massarat, Laurie Rumker, Melissa Gymrek, Soumya Raychaudhuri, Wei Zhou, and Tiffany Amariuta. 2025. “Estimating the -Heritability of Gene Expression Using Single Cell Expression Profiles Controls False Positive Rate of eGene Detection.” *bioRxiv*. https://doi.org/10.1101/2025.02.24.639892.